Mathematics Extended Essay

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Long-term Behaviour in a Multi-Level Food Chain Model

How effectively does a linear system of first-order ordinary differential equations model the long-term behaviour of a multi-trophic-level ecosystem subject to a pulse disturbance?

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Chapter 1

Introduction

Events causing organism mortality are called *ecological disturbances* [1]. These are classified as *pulse disturbances*, occurring over short timescales, or *press disturbances*, occurring over longer timescales [2], relative to the population's reproductive rate. We interpret pulse disturbances as natural disasters (e.g., wildfires) and press disturbances as persistent human-caused mortality (e.g., hunting).

Modelling such disturbances may help predict population behaviour following catastrophes. We examine the long-term effects of pulse disturbances in an ecosystem with at least two *trophic levels* (food-chain levels) through this research question:

How effectively does a linear system of first-order ordinary differential equations model the long-term behaviour of a multi-trophic-level ecosystem subject to a pulse disturbance?

By constructing and analysing a system of linear first-order ordinary differential equations (ODEs), we prove convergence to a unique equilibrium as time $t \to \infty$ following a pulse disturbance in a food chain with at least two trophic levels. We also assess the model's effectiveness in capturing real-world behaviour and the suitability of linear first-order ODEs for this task.

The essay proceeds as follows. Chapter 2 [p. 2] defines the notation, assumptions, differential equations, and parameter restrictions. Chapter 3 [p. 7] derives a unique equilibrium, uses numerical simulations to motivate a conjecture on long-term behaviour, introduces a change of variables, and proves convergence to equilibrium. Chapter 4 [p. 34] discusses and evaluates the results and model, while Chapter 5 [p. 39] concludes the essay.

Chapter 2

The Model

We introduce the model's notation, assumptions, differential equations, and parameter restrictions, which link to "linear system of first-order ordinary differential equations", "multi-trophic-level ecosystem", and "pulse disturbance" in our research question:

How effectively does a linear system of first-order ordinary differential equations model the long-term behaviour of a multi-trophic-level ecosystem subject to a pulse disturbance?

2.1 Population Indexing

Consider a food chain with k trophic levels, where $k \in \mathbb{Z}^+$ and $k \geq 2$. Let P_n denote the population at trophic level n, where $n \in \mathbb{Z}^+$ and $1 \leq n \leq k$. Here, P_1 represents the apex predator and P_k the *basal population* with no prey below it (Figure 2.1).

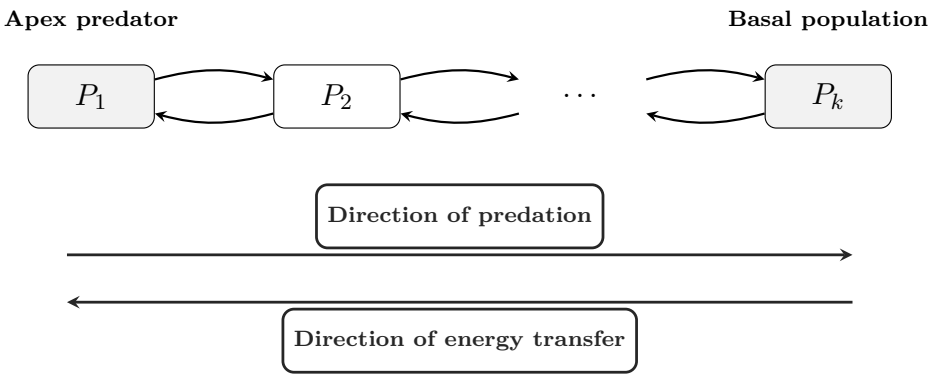


Figure 2.1: Food chain indexing notation

2.2 Assumptions

The model considers a food chain $(P_1(t), \ldots, P_k(t))$ of size k, with each population $P_n : [0, \infty) \to \mathbb{R}$. Nonnegativity conditions are omitted, as our results are independent of the populations' signs. However, assume $P_n(t) > 0$ for $t \geq 0$ and $n = 1, \ldots, k$ unless stated otherwise, as the ecological motivation fails for $P_n(t) \leq 0$. We will, however, also discuss extinct populations when necessary.

We assume:

1. One-to-one predator–prey relationships, neglecting interactions outside the chain to reduce complexity.
2. An environment with negligible genetic adaptation, as adaptation is species- and environment-specific and therefore not generalisable across food chains.
3. Geographic and age distributions are ignored to simplify the mathematics.
4. At $t = 0$, a pulse disturbance decreases the population at level n from equilibrium by $|\Delta_n|$, where $\Delta_n \leq 0$, giving

$$P_n(0) = P_n^* + \Delta_n, \qquad n = 1, \ldots, k.$$

   Here, P_n^* denotes the equilibrium population at trophic level n; we prove the existence of a unique equilibrium $(P_1^*, \ldots, P_k^*)$ in Chapter 3 [p. 7].

2.3 Differential Equations

2.3.1 System of Equations

The food chain is modelled by the following system of differential equations:

$$\frac{dP_n}{dt} = \begin{cases} \lambda_1 - \beta_1 P_1 + \gamma_1 P_1 + c_1 P_2, & n = 1 \\ \lambda_n - b_n P_{n-1} - \beta_n P_n + \gamma_n P_n + c_n P_{n+1}, & 2 \leq n \leq k-1 \\ \lambda_k - b_k P_{k-1} - \beta_k P_k + \gamma_k P_k, & n = k. \end{cases} \tag{2.1}$$

Or equivalently,

$$\frac{dP_n}{dt} = \begin{cases} \lambda_1 - (\beta_1 - \gamma_1) P_1 + c_1 P_2, & n = 1 \\ \lambda_n - b_n P_{n-1} - (\beta_n - \gamma_n) P_n + c_n P_{n+1}, & 2 \leq n \leq k-1 \\ \lambda_k - b_k P_{k-1} - (\beta_k - \gamma_k) P_k, & n = k. \end{cases} \tag{2.2}$$

We define $a_n = \beta_n - \gamma_n$. We write β_n and γ_n separately in (2.1) and (2.2) for ease of ecological interpretation in the following section.

2.3.2 Parameters

The parameter definitions and restrictions are:

- $\gamma_n > 0$ represents a species' tendency to reproduce under ideal conditions. It is positive and multiplies P_n in (2.1) because larger populations produce more offspring under such conditions.
- $\beta_n > \gamma_n > 0$ (*density-dependent regulation*) represents mortality or emigration at a rate proportional to population. It is positive and multiplies P_n in (2.1) because higher population densities increase mortality and emigration via disease and competition for resources. The restriction $\beta_n > \gamma_n$ is justified after introducing the remaining parameters needed to explain it.

- $b_n > 0,\ n > 1$ represents the effect of predator abundance on P_n. It is positive and multiplies P_{n-1} in (2.1) [p. 4] because larger predator populations decrease the prey growth rate.

- $c_n > 0,\ n < k$ represents the effect of prey abundance on P_n. It is positive and multiplies P_{n+1} in (2.1) [p. 4] because greater prey abundance increases the predator growth rate.

- λ_n in (2.1) [p. 4] represents population-independent effects.

  There are three cases for λ_n:

  1. $\lambda_n > 0$ represents a constant positive effect on the population, such as immigration or seed inflow from other ecosystems. Immigrants are unaware of the population size they are immigrating to, while seeds from other ecosystems are independent of the considered population size. Immigration is modelled this way, but emigration increases with population size and is included in β_n.
  2. $\lambda_n = 0$ represents no population-independent effect.
  3. $\lambda_n < 0$ represents a constant negative effect, such as hunting/harvesting independent of population size. However, hunting/harvesting is usually modelled as population-dependent (hunters usually hunt more when populations are larger).

  Because population-independent hunting/harvesting is unrealistic and we only consider pulse disturbances, we restrict $\lambda_n \geq 0$ for $n = 1, \ldots, k$, with the constraint $\lambda_k > 0$. The latter is justified in the next item, as explaining the sign of a_n is required first.

- $a_n = \beta_n - \gamma_n > 0$ represents the net effect of reproduction and density-dependent mortality. We now justify $\beta_n > \gamma_n$ and $\lambda_k > 0$.

  Recall the system from (2.2) [p. 4]:

$$\frac{dP_n}{dt} = \begin{cases} \lambda_1 - (\beta_1 - \gamma_1)P_1 + c_1 P_2, & n = 1 \\ \lambda_n - b_n P_{n-1} - (\beta_n - \gamma_n)P_n + c_n P_{n+1}, & 2 \leq n \leq k-1 \\ \lambda_k - b_k P_{k-1} - (\beta_k - \gamma_k)P_k, & n = k. \end{cases} \tag{2.3}$$

  Suppose that for P_n, $n \neq k$, the prey is extinct ($P_{n+1} = 0$) and $\lambda_n = 0$. Also assume no predation on P_n ($P_{n-1} = 0$) for $n \neq 1$; this is automatic for $n = 1$ since P_0 is undefined.

Under these conditions, (2.3) [p. 5] becomes

$$\frac{dP_n}{dt} = \begin{cases} -(\beta_1 - \gamma_1)P_1, & n = 1 \\ -(\beta_n - \gamma_n)P_n, & 2 \leq n \leq k-1. \\ \lambda_k - (\beta_k - \gamma_k)P_k, & n = k. \end{cases} \tag{2.4}$$

Thus, P_n, $n \neq k$, must decrease as it has no food source or external input λ_n. Hence, $a_n = (\beta_n - \gamma_n) > 0$ is required for realism when $n \neq k$.

We extend this logic to a_k in (2.4), restricting $a_k > 0$. This is mathematically useful because positive a_n coefficients provide structure to the system and enable the proof of the long-term behaviour theorem. However, without $\lambda_k > 0$, this would be unrealistic, as the basal population would lack a growth source. This is visible in (2.3) [p. 5], where the terms $-b_k P_{k-1}$ and $-(\beta_k - \gamma_k)P_k$ both contribute to a negative growth rate. We therefore model sunlight and seeds from other ecosystems with a single parameter $\lambda_k > 0$.

We have introduced the model's notation, assumptions, differential equations, and parameter restrictions. We now derive the main result on the system's long-term behaviour in Chapter 3 [p. 7].

Chapter 3

Results

This chapter explores the long-term behaviour of the system in Chapter 2 [p. 2] following a pulse disturbance, addressing the "long-term behaviour" in our research question:

How effectively does a linear system of first-order ordinary differential equations model the long-term behaviour of a multi-trophic-level ecosystem subject to a pulse disturbance?

In this chapter, we prove a unique equilibrium exists in Section 3.1. We then use numerical simulations in Section 3.2 [p. 18] to motivate the conjecture that populations converge to this equilibrium as $t \to \infty$ following a pulse disturbance. In Section 3.3 [p. 19], we rewrite the system in terms of displacements from the equilibrium, simplifying the proof since showing these displacements approach zero is easier than showing populations approach parameter-dependent equilibrium values. Finally, in Section 3.4 [p. 22], we prove the conjecture using the displacement result.

3.1 Equilibria

Definition 3.1 (Equilibrium)**.** An equilibrium of the system is a tuple

$$(P_1^*, \ldots, P_k^*),$$

such that

$$\frac{dP_n}{dt} = 0, \qquad n = 1, \ldots, k,$$

when $(P_1, \ldots, P_k) = (P_1^*, \ldots, P_k^*)$. We refer to P_n^* as the equilibrium population at trophic level n.

However, we must show that $(P_1^*, \ldots, P_k^*)$ exists. We first assume an equilibrium exists and derive its properties, showing it is uniquely determined by the system parameters. We then substitute these values to verify they satisfy the system, proving the existence and uniqueness of the equilibrium.

3.1.1 Derivation of Equilibria

We substitute the equilibrium $(P_1^*, \ldots, P_k^*)$ into the system (2.2) [p. 4] to relate the equilibrium populations to the system parameters, the first step towards expressing them in terms of these parameters.

Lemma 3.2. *If an equilibrium $(P_1^*, \ldots, P_k^*)$ exists (Definition 3.1 [p. 7]), it satisfies the following equations:*

$$P_1^* = \frac{\lambda_1 + c_1 P_2^*}{a_1}, \tag{3.1}$$

$$P_n^* = \frac{\lambda_n - b_n P_{n-1}^* + c_n P_{n+1}^*}{a_n}, \qquad 2 \leq n \leq k-1, \tag{3.2}$$

$$P_k^* = \frac{\lambda_k - b_k P_{k-1}^*}{a_k}. \tag{3.3}$$

Proof. Consider (2.2) [p. 4] with $a_n = \beta_n - \gamma_n$:

$$\frac{dP_n}{dt} = \begin{cases} \lambda_1 - a_1 P_1 + c_1 P_2, & n = 1 \\ \lambda_n - b_n P_{n-1} - a_n P_n + c_n P_{n+1}, & 2 \leq n \leq k-1, \\ \lambda_k - b_k P_{k-1} - a_k P_k, & n = k. \end{cases} \tag{3.4}$$

If an equilibrium $(P_1^*, \ldots, P_k^*)$ exists (Definition 3.1 [p. 7]), substituting it into (3.4) gives:

$$0 = \lambda_1 - a_1 P_1^* + c_1 P_2^*, \tag{3.5}$$

$$0 = \lambda_n - b_n P_{n-1}^* - a_n P_n^* + c_n P_{n+1}^*, \quad 2 \leq n \leq k-1, \tag{3.6}$$

$$0 = \lambda_k - b_k P_{k-1}^* - a_k P_k^*. \tag{3.7}$$

Rearranging (3.5) [p. 8] for P_1^*,

$$P_1^* = \frac{\lambda_1 + c_1 P_2^*}{a_1}.$$

Division by a_n is valid because $a_n \neq 0$ for $n = 1, \ldots, k$ (see Section 2.3.2 [p. 4] for parameters).

Rearranging (3.6) [p. 8] for P_n^*,

$$P_n^* = \frac{\lambda_n - b_n P_{n-1}^* + c_n P_{n+1}^*}{a_n}, \quad 2 \leq n \leq k-1. \tag{3.8}$$

Similarly, from (3.7) [p. 8]:

$$P_k^* = \frac{\lambda_k - b_k P_{k-1}^*}{a_k}.$$

□

3.1.2 Recursive Linear Structure of Equilibria

The result (3.8) in Lemma 3.2 [p. 8] is complex for intermediate equilibria ($2 \leq n \leq k-1$), as P_n^* depends on both P_{n-1}^* and P_{n+1}^*.

We thus show that each population's equilibrium can be expressed in terms of just another population's equilibrium, simplifying later results.

Lemma 3.3. *If an equilibrium $(P_1^*, \ldots, P_k^*)$ exists (Definition 3.1 [p. 7]), it satisfies*

$$Q(n): \qquad P_n^* \equiv A_n P_{n+1}^* + B_n, \qquad A_n > 0, \qquad B_n \in \mathbb{R}, \qquad 1 \leq n < k.$$

Proof. We proceed by induction on n.

Base Case

For $Q(1)$, recall (3.1) [p. 8]:

$$P_1^* = \frac{c_1 P_2^* + \lambda_1}{a_1}. \tag{3.9}$$

Let $c_1/a_1 = A_1$ and $\lambda_1/a_1 = B_1$.

Note $A_1 > 0$ because $c_1 > 0$ and $a_1 > 0$ (see Section 2.3.2 [p. 4] for parameters).

Note $B_1 \in \mathbb{R}$ because $\lambda_1 \geq 0$ and $a_1 > 0$.

Then $P_1^* = A_1 P_2^* + B_1$, $A_1 > 0$, $B_1 \in \mathbb{R}$, so $Q(1)$ is true.

Inductive Hypothesis

Assume for some $1 \leq \ell < k - 1$, $Q(\ell)$ is true, i.e.,

$$P_\ell^* = A_\ell P_{\ell+1}^* + B_\ell, \qquad A_\ell > 0, \qquad B_\ell \in \mathbb{R}. \tag{3.10}$$

We exclude $\ell = k - 1$, because the inductive step would involve $\ell + 1 = k$, which would be defined in terms of nonexistent P_{k+1}^*.

Inductive Step

We must show $Q(\ell + 1)$ follows from the assumption in (3.10), or,

$$P_{\ell+1}^* = A_{\ell+1} P_{\ell+2}^* + B_{\ell+1}, \qquad A_{\ell+1} > 0, \qquad B_{\ell+1} \in \mathbb{R}.$$

Recall (3.2) [p. 8]:

$$P_n^* = \frac{\lambda_n - b_n P_{n-1}^* + c_n P_{n+1}^*}{a_n}, \quad 2 \leq n \leq k - 1. \tag{3.11}$$

From (3.11) [p. 10], with $n = \ell + 1$,

$$P^*_{\ell+1} = \frac{-b_{\ell+1}P^*_\ell + c_{\ell+1}P^*_{\ell+2} + \lambda_{\ell+1}}{a_{\ell+1}}. \tag{3.12}$$

This holds since $\ell < k - 1$ by the inductive hypothesis, so $\ell + 1 \leq k - 1$.

Using the inductive hypothesis from (3.10) [p. 10], we substitute $P^*_\ell = A_\ell P^*_{\ell+1} + B_\ell$ into (3.12), giving

$$P^*_{\ell+1} = \frac{-b_{\ell+1}(A_\ell P^*_{\ell+1} + B_\ell) + c_{\ell+1}P^*_{\ell+2} + \lambda_{\ell+1}}{a_{\ell+1}}.$$

Solving for $P^*_{\ell+1}$,

$$P^*_{\ell+1} = \frac{-b_{\ell+1}A_\ell P^*_{\ell+1} - b_{\ell+1}B_\ell + c_{\ell+1}P^*_{\ell+2} + \lambda_{\ell+1}}{a_{\ell+1}}.$$

$$a_{\ell+1}P^*_{\ell+1} + b_{\ell+1}A_\ell P^*_{\ell+1} = c_{\ell+1}P^*_{\ell+2} + \lambda_{\ell+1} - b_{\ell+1}B_\ell.$$

$$(a_{\ell+1} + b_{\ell+1}A_\ell)P^*_{\ell+1} = c_{\ell+1}P^*_{\ell+2} + \lambda_{\ell+1} - b_{\ell+1}B_\ell. \tag{3.13}$$

Since $a_{\ell+1} > 0$, $b_{\ell+1} > 0$ (Section 2.3.2 [p. 4]), and $A_\ell > 0$ by the inductive hypothesis (3.10) [p. 10], we have $a_{\ell+1} + b_{\ell+1}A_\ell > 0$. Thus dividing (3.13) by $a_{\ell+1} + b_{\ell+1}A_\ell$ gives:

$$P^*_{\ell+1} = \frac{c_{\ell+1}}{a_{\ell+1} + b_{\ell+1}A_\ell}P^*_{\ell+2} + \frac{\lambda_{\ell+1} - b_{\ell+1}B_\ell}{a_{\ell+1} + b_{\ell+1}A_\ell}. \tag{3.14}$$

Let $c_{\ell+1}/(a_{\ell+1} + b_{\ell+1}A_\ell) = A_{\ell+1}$ and $(\lambda_{\ell+1} - b_{\ell+1}B_\ell)/(a_{\ell+1} + b_{\ell+1}A_\ell) = B_{\ell+1}$.

Note $A_{\ell+1} > 0$ because $c_{\ell+1} > 0$.

Note $B_{\ell+1} \in \mathbb{R}$ because $\lambda_{\ell+1} \geq 0$, $b_{\ell+1} > 0$, and $B_\ell \in \mathbb{R}$ by the inductive hypothesis (3.10) [p. 10]. Then,

$$P^*_{\ell+1} = A_{\ell+1}P^*_{\ell+2} + B_{\ell+1}, \qquad A_{\ell+1} > 0, \qquad B_{\ell+1} \in \mathbb{R}.$$

Thus $Q(\ell)$ implies $Q(\ell + 1)$. Since $Q(1)$ is true, $Q(n)$ is true for $1 \leq n \leq k - 1$. □

We now have a recursive definition of each equilibrium in terms of another equilibrium, but this required introducing A_n and B_n in Lemma 3.3 [p. 9]. We now show that A_n can be written in terms of A_{n-1}, which can be written in terms of A_{n-2}, and so on, so that A_n can be expressed in terms of just A_1 (and similarly B_n). Since $A_1 = c_1/a_1$ and $B_1 = \lambda_1/a_1$, showing this establishes that A_n and B_n are determined by the system parameters. This is needed because we later express P_k^* in terms of A_{k-1} and B_{k-1}; to show that if P_k^* exists, it is uniquely determined by the system parameters, we must show that A_{k-1} and B_{k-1} are as well. We proved this in Lemma 3.3 [p. 9], but state it separately for clarity, rather than combining both results in one proof.

Lemma 3.4. *For $2 \leq n \leq k-1$, A_n and B_n in Lemma 3.3 [p. 9] are recursively defined by*

$$A_n = \frac{c_n}{a_n + b_n A_{n-1}}, \quad B_n = \frac{\lambda_n - b_n B_{n-1}}{a_n + b_n A_{n-1}}.$$

Proof. We proved in Lemma 3.3 [p. 9] that if

$$P_\ell^* = A_\ell P_{\ell+1}^* + B_\ell, \qquad A_\ell > 0, \qquad B_\ell \in \mathbb{R} \tag{3.15}$$

is true for some $1 \leq \ell < k-1$, we have (3.14) [p. 11]:

$$P_{\ell+1}^* = \frac{c_{\ell+1}}{a_{\ell+1} + b_{\ell+1} A_\ell} P_{\ell+2}^* + \frac{\lambda_{\ell+1} - b_{\ell+1} B_\ell}{a_{\ell+1} + b_{\ell+1} A_\ell}. \tag{3.16}$$

Since we showed $Q(1)$ is true in Lemma 3.3 [p. 9], defining $A_1 = c_1/a_1$ and $B_1 = \lambda_1/a_1$, $Q(2), \ldots, Q(k-1)$ are true.

So $A_n = c_n/(a_n + b_n A_{n-1})$ and $B_n = (\lambda_n - b_n B_{n-1})/(a_n + b_n A_{n-1})$ for $2 \leq n \leq k-1$ by (3.16) with $A_1 = c_1/a_1$ and $B_1 = \lambda_1/a_1$.

□

3.1.3 Explicit Form of P_k^*

Here, we show that if an equilibrium exists, P_k^* is uniquely determined by the system parameters.

This establishes equilibrium uniqueness because Lemma 3.3 [p. 9] shows that

$$P_n^* \equiv A_n P_{n+1}^* + B_n, \qquad n = 1, \ldots, k-1,$$

while Lemma 3.4 [p. 12] shows that A_n and B_n are determined by the system parameters. Thus, if P_k^* is uniquely determined by these parameters, the recursive definitions uniquely determine all other equilibria.

Lemma 3.5. *If an equilibrium $(P_1^*, \ldots, P_k^*)$ exists (Definition 3.1 [p. 7]), then*

$$P_k^* = \frac{\lambda_k - b_k B_{k-1}}{a_k + b_k A_{k-1}},$$

with

$$A_n = \frac{c_n}{a_n + b_n A_{n-1}}, \quad B_n = \frac{\lambda_n - b_n B_{n-1}}{a_n + b_n A_{n-1}}$$

for $2 \leq n \leq k-1$ with $A_1 = c_1/a_1$ and $B_1 = \lambda_1/a_1$.

Proof. If P_k^* and P_{k-1}^* exist, they satisfy Lemma 3.2 [p. 8],

$$P_k^* = \frac{\lambda_k - b_k P_{k-1}^*}{a_k}. \tag{3.17}$$

They must also satisfy Lemma 3.3 [p. 9],

$$P_{k-1}^* = A_{k-1} P_k^* + B_{k-1} \qquad A_{k-1} > 0, \qquad B_{k-1} \in \mathbb{R}. \tag{3.18}$$

Substituting (3.18) into (3.17) gives

$$P_k^* = \frac{\lambda_k - b_k (A_{k-1} P_k^* + B_{k-1})}{a_k}.$$

Solving for P_k^*,

$$P_k^* = \frac{\lambda_k - b_k A_{k-1} P_k^* - b_k B_{k-1}}{a_k}.$$

$$a_k P_k^* = \lambda_k - b_k A_{k-1} P_k^* - b_k B_{k-1}.$$

$$a_k P_k^* + b_k A_{k-1} P_k^* = \lambda_k - b_k B_{k-1}.$$

$$(a_k + b_k A_{k-1}) P_k^* = \lambda_k - b_k B_{k-1}. \tag{3.19}$$

Since $a_k > 0$, $b_k > 0$ (see Section 2.3.2 [p. 4] for parameters), and $A_{k-1} > 0$ (Lemma 3.3 [p. 9]), $a_k + b_k A_{k-1} \neq 0$.

Thus, dividing (3.19) by $a_k + b_k A_{k-1}$,

$$P_k^* = \frac{\lambda_k - b_k B_{k-1}}{a_k + b_k A_{k-1}}. \tag{3.20}$$

□

Thus, if an equilibrium exists, it is uniquely determined by the system parameters. We can now construct a candidate equilibrium through our recursive definitions. If this candidate satisfies the system of equations, then an equilibrium exists and is unique.

3.1.4 Proof of Equilibrium

Theorem 3.6. *The system (3.4) [p. 8]*

$$\frac{dP_n}{dt} = \begin{cases} \lambda_1 - a_1 P_1 + c_1 P_2, & n = 1 \\ \lambda_n - b_n P_{n-1} - a_n P_n + c_n P_{n+1}, & 2 \leq n \leq k-1, \\ \lambda_k - b_k P_{k-1} - a_k P_k, & n = k. \end{cases} \tag{3.21}$$

has a unique equilibrium (Definition 3.1 [p. 7]).

Proof. Let:

$$\tilde{P}_k = \frac{\lambda_k - b_k B_{k-1}}{a_k + b_k A_{k-1}}. \tag{3.22}$$

Then for $n = k-1, k-2, \ldots, 1$, define:

$$\tilde{P}_n = A_n \tilde{P}_{n+1} + B_n. \tag{3.23}$$

where A_n, B_n are defined by:

$$A_1 = \frac{c_1}{a_1}. \tag{3.24}$$

$$B_1 = \frac{\lambda_1}{a_1}. \tag{3.25}$$

$$A_n = \frac{c_n}{a_n + b_n A_{n-1}}, \qquad 2 \le n \le k-1. \tag{3.26}$$

$$B_n = \frac{\lambda_n - b_n B_{n-1}}{a_n + b_n A_{n-1}}, \qquad 2 \le n \le k-1. \tag{3.27}$$

We proved in Lemma 3.3 [p. 9], Lemma 3.4 [p. 12], and Lemma 3.5 [p. 13] that, if an equilibrium exists, these are valid expressions for it.

We use different notation, $(\tilde{P}_1, \ldots, \tilde{P}_k)$, to signify constructed candidate values; $(P_1^*, \ldots, P_k^*)$ was used assuming an equilibrium exists.

We now verify that $(\tilde{P}_1, \ldots, \tilde{P}_k)$ satisfies the equilibrium equations:

$$V_1 = \lambda_1 - a_1\tilde{P}_1 + c_1\tilde{P}_2 = 0, \qquad \text{from (3.5) [p. 8]}, \tag{3.28}$$

$$V_2 = \lambda_n - b_n\tilde{P}_{n-1} - a_n\tilde{P}_n + c_n\tilde{P}_{n+1} = 0, \qquad 2 \le n \le k-1, \tag{3.29}$$

from (3.6) [p. 8].

$$V_3 = \lambda_k - b_k\tilde{P}_{k-1} - a_k\tilde{P}_k = 0, \qquad \text{from (3.7) [p. 8]}. \tag{3.30}$$

Apex Equation: We substitute $\tilde{P}_1 = A_1\tilde{P}_2 + B_1$ from (3.23) [p. 14], $A_1 = c_1/a_1$ from (3.24), and $B_1 = \lambda_1/a_1$ from (3.25) into (3.28):

$$\begin{aligned} V_1 &= \lambda_1 - a_1\tilde{P}_1 + c_1\tilde{P}_2 \\ &= \lambda_1 - a_1\left(A_1\tilde{P}_2 + B_1\right) + c_1\tilde{P}_2 \\ &= \lambda_1 - a_1\left(\frac{c_1}{a_1}\tilde{P}_2 + \frac{\lambda_1}{a_1}\right) + c_1\tilde{P}_2 \\ &= 0. \end{aligned}$$

Thus, $\tilde{P}_1$ satisfies the apex equation (3.28) [p. 15].

Intermediate Equations ($2 \leq n \leq k-1$): We show (3.29) [p. 15],

$$V_2 = \lambda_n - b_n \tilde{P}_{n-1} - a_n \tilde{P}_n + c_n \tilde{P}_{n+1} = 0, \tag{3.31}$$

is satisfied by $\tilde{P}_n$ for $2 \leq n \leq k-1$.

Substituting $\tilde{P}_{n-1} = A_{n-1}\tilde{P}_n + B_{n-1}$ and $\tilde{P}_n = A_n \tilde{P}_{n+1} + B_n$ from (3.23) [p. 14] into (3.31) and rearranging,

$$V_2 = \lambda_n - b_n(A_{n-1}\tilde{P}_n + B_{n-1}) - a_n(A_n \tilde{P}_{n+1} + B_n) + c_n \tilde{P}_{n+1}.$$

$$V_2 = \lambda_n - b_n B_{n-1} - a_n B_n - b_n A_{n-1}\tilde{P}_n + (-a_n A_n + c_n)\tilde{P}_{n+1}.$$

$$V_2 = (\lambda_n - b_n B_{n-1} - a_n B_n) - b_n A_{n-1}\tilde{P}_n + (-a_n A_n + c_n)\tilde{P}_{n+1}. \tag{3.32}$$

From the definitions of A_n and B_n in (3.26) [p. 15] and (3.27) [p. 15],

$$A_n = \frac{c_n}{a_n + b_n A_{n-1}}. \tag{3.33}$$

$$B_n = \frac{\lambda_n - b_n B_{n-1}}{a_n + b_n A_{n-1}}. \tag{3.34}$$

From (3.33),

$$(a_n + b_n A_{n-1})A_n = c_n.$$

$$a_n A_n + b_n A_{n-1} A_n = c_n.$$

$$c_n - a_n A_n = b_n A_{n-1} A_n. \tag{3.35}$$

From (3.34),

$$(a_n + b_n A_{n-1})B_n = \lambda_n - b_n B_{n-1}.$$

$$a_n B_n + b_n A_{n-1} B_n = \lambda_n - b_n B_{n-1}.$$

$$\lambda_n - b_n B_{n-1} - a_n B_n = b_n A_{n-1} B_n. \tag{3.36}$$

Recall (3.32) [p. 16]:

$$V_2 = (\lambda_n - b_n B_{n-1} - a_n B_n) - b_n A_{n-1} \tilde{P}_n + (-a_n A_n + c_n) \tilde{P}_{n+1}. \tag{3.37}$$

Substituting (3.35) [p. 16] and (3.36) into (3.37),

$$V_2 = b_n A_{n-1} B_n - b_n A_{n-1} \tilde{P}_n + b_n A_{n-1} A_n \tilde{P}_{n+1}, \tag{3.38}$$

Factoring $b_n A_{n-1}$ from (3.38),

$$V_2 = b_n A_{n-1} (B_n - \tilde{P}_n + A_n \tilde{P}_{n+1}).$$

But $\tilde{P}_n = A_n \tilde{P}_{n+1} + B_n$ from (3.23) [p. 14], so $B_n - \tilde{P}_n + A_n \tilde{P}_{n+1} = 0$.

Therefore:

$$V_2 = \lambda_n - b_n \tilde{P}_{n-1} - a_n \tilde{P}_n + c_n \tilde{P}_{n+1} = 0.$$

Hence, $\tilde{P}_n$ satisfies the intermediate equation (3.29) [p. 15] for $2 \leq n \leq k-1$.

Basal Equation:

We show that $\tilde{P}_k$ satisfies (3.30) [p. 15]:

$$V_3 = \lambda_k - b_k \tilde{P}_{k-1} - a_k \tilde{P}_k = 0. \tag{3.39}$$

Substituting $\tilde{P}_{k-1} = A_{k-1} \tilde{P}_k + B_{k-1}$ from (3.23) [p. 14] into (3.39),

$$V_3 = \lambda_k - b_k (A_{k-1} \tilde{P}_k + B_{k-1}) - a_k \tilde{P}_k,$$

$$V_3 = \lambda_k - b_k B_{k-1} - (a_k + b_k A_{k-1}) \tilde{P}_k. \tag{3.40}$$

But from (3.22) [p. 14], $\tilde{P}_k = \dfrac{\lambda_k - b_k B_{k-1}}{a_k + b_k A_{k-1}}$. Substituting this into (3.40) [p. 17],

$$V_3 = \lambda_k - b_k B_{k-1} - (a_k + b_k A_{k-1}) \left(\frac{\lambda_k - b_k B_{k-1}}{a_k + b_k A_{k-1}} \right),$$

so

$$V_3 = \lambda_k - b_k B_{k-1} - (\lambda_k - b_k B_{k-1}) = 0.$$

Hence, $\tilde{P}_k$ satisfies the basal equation (3.30) [p. 15].

All three equilibrium equations hold, so $(\tilde{P}_1, \ldots, \tilde{P}_k)$ satisfies the equilibrium equations.

Since we proved that an equilibrium must be of this form and that this form satisfies the equilibrium equations, existence and uniqueness have been proven. □

3.2 Conjecture via Numerical Simulations

When $k > 2$, the system cannot be solved simply, so we investigate it numerically to formulate a conjecture which we prove. We proved in Theorem 3.6 [p. 14] that a unique equilibrium exists and is determined by the system parameters. Thus, in Figure 3.1, we compute the equilibria and draw them as dashed lines, with each colour corresponding to the respective population.

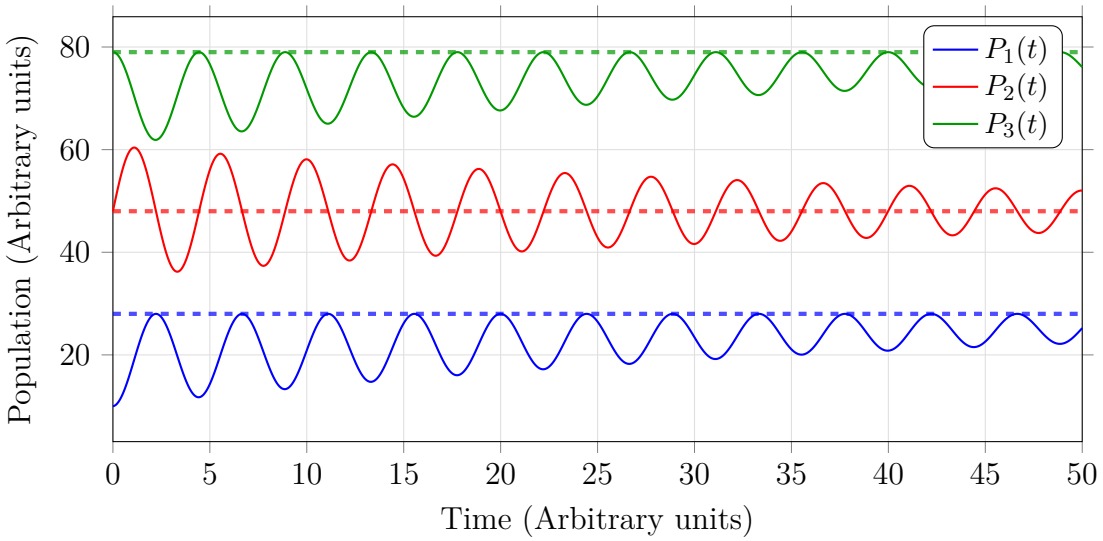


Figure 3.1: Populations appearing to return to equilibrium.

All populations appear to converge to equilibrium (Figure 3.1 [p. 18]). We therefore conjecture that

$$\lim_{t\to\infty} P_n(t) = P_n^*, \qquad 1 \le n \le k.$$

We now prove the conjecture.

3.3 Conversion of Populations to Displacements

Since the system has a unique equilibrium (Theorem 3.6 [p. 14]), we rewrite the system in terms of displacements from equilibrium. As noted in Section 3.1 [p. 7], it is easier to show that the displacements approach zero as $t \to \infty$ than that the populations approach the equilibrium values determined by the system parameters.

Definition 3.7. Define displacement $s_n(t)$ as:

$$s_n(t) = P_n(t) - P_n^*,$$

for $1 \le n \le k$, where P_n^* is the equilibrium value of P_n (Definition 3.1 [p. 7]) and P_n is the population at time t by system (3.4) [p. 8].

Theorem 3.8. *Given the system (3.4) [p. 8] or*

$$\frac{dP_n}{dt} = \begin{cases} \lambda_1 - a_1P_1 + c_1P_2, & n = 1 \\ \lambda_n - b_nP_{n-1} - a_nP_n + c_nP_{n+1}, & 2 \le n \le k-1, \\ \lambda_k - b_kP_{k-1} - a_kP_k, & n = k, \end{cases} \tag{3.41}$$

and $s_n(t) = P_n(t) - P_n^*$ (Definition 3.7), the system can be rewritten in terms of $s_n(t)$ as:

$$\frac{ds_n}{dt} = \begin{cases} -a_1s_1 + c_1s_2, & n = 1, \\ -b_ns_{n-1} - a_ns_n + c_ns_{n+1}, & 2 \le n \le k-1, \\ -b_ks_{k-1} - a_ks_k, & n = k. \end{cases} \tag{3.42}$$

Proof. Since P_n^* is constant, differentiating $s_n(t) = P_n(t) - P_n^*$ gives

$$\frac{ds_n}{dt} = \frac{dP_n}{dt}.$$

Therefore from (3.41) [p. 19],

$$\frac{ds_n}{dt} = \frac{dP_n}{dt} = \begin{cases} \lambda_1 - a_1 P_1 + c_1 P_2, & n = 1 \\ \lambda_n - b_n P_{n-1} - a_n P_n + c_n P_{n+1}, & 2 \leq n \leq k-1, \\ \lambda_k - b_k P_{k-1} - a_k P_k, & n = k. \end{cases} \tag{3.43}$$

We now substitute $P_n = s_n + P_n^*$ (Definition 3.7) into (3.43). We have omitted function notation as this is typical with differential equations.

Case $n = 1$:

From (3.43),

$$\frac{ds_1}{dt} = \lambda_1 - a_1 P_1 + c_1 P_2. \tag{3.44}$$

Substituting $P_n = s_n + P_n^*$ (Definition 3.7 [p. 19]) into (3.44),

$$\frac{ds_1}{dt} = \lambda_1 - a_1(s_1 + P_1^*) + c_1(s_2 + P_2^*).$$

Rearranging,

$$\frac{ds_1}{dt} = (\lambda_1 - a_1 P_1^* + c_1 P_2^*) - a_1 s_1 + c_1 s_2.$$

We proved in Lemma 3.2 [p. 8] that if an equilibrium exists, we have (3.5) [p. 8]:

$$\lambda_1 - a_1 P_1^* + c_1 P_2^* = 0,$$

so

$$\frac{ds_1}{dt} = -a_1 s_1 + c_1 s_2. \tag{3.45}$$

Case $2 \leq n \leq k-1$:

From (3.43):

$$\frac{ds_n}{dt} = \lambda_n - b_n P_{n-1} - a_n P_n + c_n P_{n+1}. \tag{3.46}$$

Substituting $P_n = s_n + P_n^*$ (Definition 3.7 [p. 19]) into (3.46) [p. 20],

$$\frac{ds_n}{dt} = \lambda_n - b_n(s_{n-1} + P_{n-1}^*) - a_n(s_n + P_n^*) + c_n(s_{n+1} + P_{n+1}^*).$$

Rearranging,

$$\frac{ds_n}{dt} = \left(\lambda_n - b_n P_{n-1}^* - a_n P_n^* + c_n P_{n+1}^*\right) - b_n s_{n-1} - a_n s_n + c_n s_{n+1}. \tag{3.47}$$

We proved in Lemma 3.2 [p. 8] that if an equilibrium exists then we have (3.6) [p. 8]:

$$\lambda_n - b_n P_{n-1}^* - a_n P_n^* + c_n P_{n+1}^* = 0. \tag{3.48}$$

Substituting (3.48) into (3.47):

$$\frac{ds_n}{dt} = -b_n s_{n-1} - a_n s_n + c_n s_{n+1}. \tag{3.49}$$

Case $n = k$:

From (3.43) [p. 20]:

$$\frac{ds_k}{dt} = \lambda_k - b_k P_{k-1} - a_k P_k. \tag{3.50}$$

Substituting $P_n = s_n + P_n^*$ (Definition 3.7 [p. 19]) into (3.50),

$$\frac{ds_k}{dt} = \lambda_k - b_k(s_{k-1} + P_{k-1}^*) - a_k(s_k + P_k^*).$$

Rearranging,

$$\frac{ds_k}{dt} = \left(\lambda_k - b_k P_{k-1}^* - a_k P_k^*\right) - b_k s_{k-1} - a_k s_k. \tag{3.51}$$

We proved in Lemma 3.2 [p. 8] that if an equilibrium exists we have (3.7) [p. 8]:

$$\lambda_k - b_k P_{k-1}^* - a_k P_k^* = 0. \tag{3.52}$$

Substituting (3.52) [p. 21] into (3.51) [p. 21],

$$\frac{ds_k}{dt} = -b_k s_{k-1} - a_k s_k. \tag{3.53}$$

Combining (3.45) [p. 20], (3.49) [p. 21], and (3.53) yields

$$\frac{ds_n}{dt} = \begin{cases} -a_1 s_1 + c_1 s_2, & n = 1, \\ -b_n s_{n-1} - a_n s_n + c_n s_{n+1}, & 2 \leq n \leq k-1, \\ -b_k s_{k-1} - a_k s_k, & n = k. \end{cases}$$

□

3.4 Convergence of Populations to Equilibrium

We now prove the conjecture from Section 3.2 [p. 18] that populations converge to equilibrium as $t \to \infty$.

Recall the system of displacements (3.42) [p. 19]:

$$\frac{ds_n}{dt} = \begin{cases} -a_1 s_1 + c_1 s_2, & n = 1, \\ -b_n s_{n-1} - a_n s_n + c_n s_{n+1}, & 2 \leq n \leq k-1, \\ -b_k s_{k-1} - a_k s_k, & n = k. \end{cases} \tag{3.54}$$

Using (3.54), define

$$f(t) = \sum_{n=1}^{k} \mu_n [s_n(t)]^2 \tag{3.55}$$

for $t \geq 0$, where $\mu_n > 0$ for $1 \leq n \leq k$. Its domain is $t \geq 0$ because in Definition 3.7 [p. 19], we defined

$$s_n(t) = P_n(t) - P_n^*, \tag{3.56}$$

where $P_n^* \in \mathbb{R}$. When introducing the model, we restricted the domain of $P_n(t)$ to $t \geq 0$ (Section 2.2 [p. 3]), so the domain of $s_n(t)$ is also $t \geq 0$.

We restrict $\mu_n > 0$ for $1 \leq n \leq k$ to exploit two properties of $f(t)$:

1. $f(t) \geq 0$ for $t \geq 0$.
2. $f(t) = 0$ if and only if $s_n(t) = 0$ for all $1 \leq n \leq k$.

These properties hold because every term in (3.55) [p. 22] is nonnegative: the terms are squared and $\mu_n > 0$ for $1 \leq n \leq k$. Thus, if $f(t) \to 0$ as $t \to \infty$, then $s_n(t) \to 0$ for all $1 \leq n \leq k$, meaning by $s_n(t) = P_n(t) - P_n^*$ from (3.56) [p. 22] that $P_n(t) \to P_n^*$.

Remark 3.9. *We define $\mu_1 = 1$ and $\mu_{n+1} = \mu_n c_n / b_{n+1}$ for $1 \leq n \leq k-1$. Since $b_{n+1}, c_n > 0$ for $1 \leq n < k$ (Section 2.3.2 [p. 4]), we have $\mu_n > 0$ for $1 \leq n \leq k$, enabling the two properties of $f(t)$ above. The usefulness of this choice becomes apparent in Lemma 3.10, where it enables cancellation in a complicated expression.*

We prove properties of $f(t)$ that show $f(t) \to 0$ as $t \to \infty$. In the following lemma (Lemma 3.10), we simplify $f'(t)$, allowing us to show that $f(t)$ is non-increasing, which helps establish $f(t) \to 0$.

Lemma 3.10. *Given (3.55) [p. 22]:*

$$f(t) = \sum_{n=1}^{k} \mu_n [s_n(t)]^2, \tag{3.57}$$

where $s_n(t) = P_n(t) - P_n^*$ (Definition 3.7 [p. 19]) and $\mu_1 = 1$ and $\mu_{n+1} = \mu_n c_n / b_{n+1}$ for $1 \leq n \leq k-1$ (Remark 3.9),

we have

$$f'(t) = -2 \sum_{n=1}^{k} \mu_n a_n [s_n(t)]^2. \tag{3.58}$$

Proof. Differentiating (3.57),

$$f'(t) = 2 \sum_{n=1}^{k} \mu_n s_n(t) \frac{ds_n}{dt}. \tag{3.59}$$

The first term of (3.59) with $n = 1$ is (excluding the factor of 2)

$$\mu_1 s_1(t) \frac{ds_1}{dt}. \tag{3.60}$$

We omit the function notation for $s_n(t)$ henceforth to simplify notation.

From (3.54) [p. 22],

$$\frac{ds_1}{dt} = -a_1 s_1 + c_1 s_2 \tag{3.61}$$

Substituting (3.61) into (3.60) [p. 23],

$$\mu_1 s_1(-a_1 s_1 + c_1 s_2) = -\mu_1 a_1 s_1^2 + \mu_1 c_1 s_1 s_2. \tag{3.62}$$

The terms of (3.59) [p. 23] for $2 \leq n \leq k-1$ are (excluding the factor of 2)

$$\sum_{n=2}^{k-1} \mu_n s_n \frac{ds_n}{dt}. \tag{3.63}$$

Recall (3.54) [p. 22] for $2 \leq n \leq k-1$:

$$\frac{ds_n}{dt} = -b_n s_{n-1} - a_n s_n + c_n s_{n+1}. \tag{3.64}$$

Substituting (3.64) into (3.63),

$$\sum_{n=2}^{k-1} \mu_n s_n(-b_n s_{n-1} - a_n s_n + c_n s_{n+1}) = -\sum_{n=2}^{k-1} \mu_n b_n s_n s_{n-1} - \sum_{n=2}^{k-1} \mu_n a_n s_n^2 + \sum_{n=2}^{k-1} \mu_n c_n s_n s_{n+1}. \tag{3.65}$$

The last term of (3.59) [p. 23] with $n = k$ is (excluding the factor of 2)

$$\mu_k s_k \frac{ds_k}{dt}. \tag{3.66}$$

Recall (3.54) [p. 22]:

$$\frac{ds_k}{dt} = -b_k s_{k-1} - a_k s_k. \tag{3.67}$$

Substituting (3.67) into (3.66),

$$\mu_k s_k(-b_k s_{k-1} - a_k s_k) = -\mu_k a_k s_k^2 - \mu_k b_k s_k s_{k-1}. \tag{3.68}$$

Hence, from (3.62) [p. 24],

$$\underbrace{-\mu_1 a_1 s_1^2}_{a_n \text{ term}} + \underbrace{\mu_1 c_1 s_1 s_2}_{c_n \text{ term}}. \tag{3.69}$$

From (3.65) [p. 24],

$$\underbrace{-\sum_{n=2}^{k-1} \mu_n b_n s_n s_{n-1}}_{b_n \text{ terms}} \underbrace{-\sum_{n=2}^{k-1} \mu_n a_n s_n^2}_{a_n \text{ terms}} + \underbrace{\sum_{n=2}^{k-1} \mu_n c_n s_n s_{n+1}}_{c_n \text{ terms}}. \tag{3.70}$$

From (3.68) [p. 24],

$$\underbrace{-\mu_k a_k s_k^2}_{a_n \text{ term}} - \underbrace{\mu_k b_k s_k s_{k-1}}_{b_n \text{ term}}. \tag{3.71}$$

Combining the a_n terms from (3.69), (3.70), and (3.71),

$$-\mu_1 a_1 s_1^2 + \sum_{n=2}^{k-1} -\mu_n a_n s_n^2 - \mu_k a_k s_k^2 = -\sum_{n=1}^{k} \mu_n a_n s_n^2. \tag{3.72}$$

Combining the b_n terms from (3.70) and (3.71),

$$\sum_{n=2}^{k-1} -\mu_n b_n s_n s_{n-1} - \mu_k b_k s_k s_{k-1} = -\sum_{n=2}^{k} \mu_n b_n s_n s_{n-1}. \tag{3.73}$$

Combining the c_n terms from (3.69) and (3.70),

$$\sum_{n=2}^{k-1} \mu_n c_n s_n s_{n+1} + \mu_1 c_1 s_1 s_2 = \sum_{n=1}^{k-1} \mu_n c_n s_n s_{n+1}. \tag{3.74}$$

Combining (3.72), (3.73), and (3.74):

$$-\sum_{n=1}^{k} \mu_n a_n s_n^2 - \sum_{n=2}^{k} \mu_n b_n s_n s_{n-1} + \sum_{n=1}^{k-1} \mu_n c_n s_n s_{n+1}. \tag{3.75}$$

The expression in (3.75) [p. 25] equals the sum below:

$$\frac{1}{2}f'(t) = \sum_{n=1}^{k} \mu_n s_n \frac{ds_n}{dt}, \tag{3.76}$$

Hence, from (3.76),

$$f'(t) = 2\left(-\sum_{n=1}^{k} \mu_n a_n s_n^2 - \sum_{n=2}^{k} \mu_n b_n s_n s_{n-1} + \sum_{n=1}^{k-1} \mu_n c_n s_n s_{n+1} \right). \tag{3.77}$$

Define

$$g(t) = -\sum_{n=2}^{k} \mu_n b_n s_n s_{n-1} + \sum_{n=1}^{k-1} \mu_n c_n s_n s_{n+1}. \tag{3.78}$$

So,

$$f'(t) = 2\left(-\sum_{n=1}^{k} \mu_n a_n s_n^2 + g(t) \right). \tag{3.79}$$

Define $m = n - 1$. Then $n = m + 1$. When $n = 2$, $m = 1$, and when $n = k$, $m = k - 1$. Therefore, we write the first term in (3.78) as

$$-\sum_{n=2}^{k} \mu_n b_n s_n s_{n-1} = -\sum_{m=1}^{k-1} \mu_{m+1} b_{m+1} s_{m+1} s_m. \tag{3.80}$$

Since m is an arbitrary variable, we rename it n. Substituting (3.80) into (3.78),

$$g(t) = -\sum_{n=1}^{k-1} \mu_{n+1} b_{n+1} s_{n+1} s_n + \sum_{n=1}^{k-1} \mu_n c_n s_n s_{n+1}. \tag{3.81}$$

We thus write (3.81) as

$$g(t) = \sum_{n=1}^{k-1} (\mu_n c_n - \mu_{n+1} b_{n+1}) s_n s_{n+1}. \tag{3.82}$$

Recall from Remark 3.9 [p. 23], we have $\mu_1 = 1$ and $\mu_{n+1} = \mu_n c_n / b_{n+1}$ for $1 \leq n \leq k-1$. Substituting this definition of μ_{n+1} into (3.82) [p. 26],

$$g(t) = \sum_{n=1}^{k-1} (\mu_n c_n - \mu_n \frac{c_n}{b_{n+1}} b_{n+1}) s_n s_{n+1} = 0.$$

We now see the usefulness of the definition of μ_n outlined in Remark 3.9 [p. 23]: it is chosen to obtain $g(t) = 0$, drastically simplifying $f'(t)$.

Recall the equation for $f'(t)$ from (3.79) [p. 26]:

$$f'(t) = 2\left(-\sum_{n=1}^{k} \mu_n a_n s_n^2 + g(t)\right). \tag{3.83}$$

Thus, substituting $g(t) = 0$ into (3.83),

$$f'(t) = -2\sum_{n=1}^{k} \mu_n a_n s_n^2. \tag{3.84}$$

□

Eq. (3.84) already shows that $f'(t) \leq 0$ for $t \geq 0$ because $\mu_n > 0$ and $a_n > 0$ for $1 \leq n \leq k$ (Section 2.3.2 [p. 4]). However, this is not sufficient to conclude that $f(t) \to 0$ as $t \to \infty$ because a non-increasing function bounded from below by 0 may converge to a positive limit. We will use Lemma 3.10 [p. 23] to form an inequality relating $f'(t)$ to $f(t)$. This is useful because to finalise the proof, we will examine a function $h(t) = e^{pt} f(t)$, where $p > 0$. Differentiating $h(t)$ gives $h'(t) = e^{pt}(f'(t) + pf(t))$. If $f'(t) \leq -pf(t)$ (we will show this for our system in the following lemma), then we conclude that $h'(t) \leq 0$; it follows that $f(t) \to 0$ as $t \to \infty$, but this will be seen in Theorem 3.12 [p. 29].

Lemma 3.11. *Given*

$$f(t) = \sum_{n=1}^{k} \mu_n s_n^2, \tag{3.85}$$

where $s_n = P_n - P_n^*$ (Definition 3.7 [p. 19]) and $\mu_1 = 1$ and $\mu_{n+1} = \mu_n c_n / b_{n+1}$ for $1 \leq n \leq k-1$ (Remark 3.9 [p. 23]), we have

$$f'(t) \leq -2a_{\min} f(t), \tag{3.86}$$

where

$$a_{\min} = \min(a_1, \ldots, a_k). \tag{3.87}$$

Proof. Since $a_n > 0$ for $1 \leq n \leq k$ (Section 2.3.2 [p. 4]),

$$a_{\min} > 0. \tag{3.88}$$

Then

$$a_n \geq a_{\min} \tag{3.89}$$

for $1 \leq n \leq k$. Recall our definition of μ_n, $\mu_1 = 1$ and $\mu_{n+1} = \mu_n c_n / b_{n+1}$ for $1 \leq n \leq k-1$ (Remark 3.9 [p. 23]). Since $b_{n+1}, c_n > 0$ for $1 \leq n < k$ (see Section 2.3.2 [p. 4]), we know that $\mu_n > 0$ for $1 \leq n \leq k$. Thus, multiplying both sides of (3.89) by the nonnegative quantity $\mu_n s_n^2$ gives

$$\mu_n a_n s_n^2 \geq a_{\min} \mu_n s_n^2.$$

Summing over $n = 1, \ldots, k$,

$$\sum_{n=1}^{k} \mu_n a_n s_n^2 \geq \sum_{n=1}^{k} a_{\min} \mu_n s_n^2.$$

Or,

$$\sum_{n=1}^{k} \mu_n a_n s_n^2 \geq a_{\min} \sum_{n=1}^{k} \mu_n s_n^2. \tag{3.90}$$

Recall the definition of $f(t)$ from (3.55) [p. 22]:

$$f(t) = \sum_{n=1}^{k} \mu_n s_n^2. \tag{3.91}$$

We substitute (3.91) [p. 28] into the right-hand side of (3.90) [p. 28],

$$\sum_{n=1}^{k} \mu_n a_n s_n^2 \geq a_{\min} f(t). \tag{3.92}$$

Recall from Lemma 3.10 [p. 23], that

$$f'(t) = -2 \sum_{n=1}^{k} \mu_n a_n s_n^2. \tag{3.93}$$

Combining (3.93) with (3.92),

$$f'(t) = -2 \sum_{n=1}^{k} \mu_n a_n s_n^2 \leq -2 a_{\min} f(t),$$

so

$$f'(t) \leq -2 a_{\min} f(t). \tag{3.94}$$

□

We now use the result (3.94) in Lemma 3.11 [p. 27] to prove that the populations converge to equilibrium as $t \to \infty$.

Theorem 3.12. *Under the model (3.4) [p. 8],*

$$\frac{dP_n}{dt} = \begin{cases} \lambda_1 - a_1 P_1 + c_1 P_2, & n = 1 \\ \lambda_n - b_n P_{n-1} - a_n P_n + c_n P_{n+1}, & 2 \leq n \leq k-1, \\ \lambda_k - b_k P_{k-1} - a_k P_k, & n = k, \end{cases} \tag{3.95}$$

and the equation (Definition 3.7 [p. 19])

$$s_n(t) = P_n(t) - P_n^*, \tag{3.96}$$

where $P_1^*, \ldots, P_k^*$ are the equilibrium values of (3.4) [p. 8] (Definition 3.1 [p. 7]), we have

$$\lim_{t\to\infty} P_n(t) = P_n^* \tag{3.97}$$

for $1 \leq n \leq k$.

Proof. With (3.55) [p. 22],

$$f(t) = \sum_{n=1}^{k} \mu_n s_n^2 \tag{3.98}$$

for $t \geq 0$, with $\mu_1 = 1$ and $\mu_{n+1} = \mu_n c_n / b_{n+1}$ for $1 \leq n \leq k-1$, let us define

$$h(t) = f(t) e^{2a_{\min} t}, \tag{3.99}$$

where

$$a_{\min} = \min(a_1, \ldots, a_k) > 0. \tag{3.100}$$

Note that $a_{\min} > 0$ because $a_n > 0$ for $1 \leq n \leq k$ (Section 2.3.2 [p. 4]). Also note that the domain of $h(t)$ is $t \geq 0$ because we defined $f(t)$ to have domain $t \geq 0$ in (3.55) [p. 22].

Differentiating both sides of (3.99),

$$h'(t) = f'(t) e^{2a_{\min} t} + 2a_{\min} f(t) e^{2a_{\min} t}. \tag{3.101}$$

Factoring $e^{2a_{\min} t}$,

$$h'(t) = e^{2a_{\min} t} \left[f'(t) + 2a_{\min} f(t) \right]. \tag{3.102}$$

From Lemma 3.11 [p. 27], we have $f'(t) + 2a_{\min} f(t) \leq 0$. Therefore, from (3.102), we conclude that

$$h'(t) \leq 0. \tag{3.103}$$

This reveals one reason we chose the exponential function $e^{2a_{\min} t}$ in (3.99); its derivative has $e^{2a_{\min} t}$ as a factor, meaning the sign of $h'(t)$ is determined by the relationship between $f'(t)$ and $f(t)$, which we derived in Lemma 3.11 [p. 27].

Eq. (3.103) [p. 30] implies that $h(t)$ is a non-increasing function. Therefore, because its domain is $t \geq 0$, we know that

$$h(t) \leq h(0), \tag{3.104}$$

for $t \geq 0$.

Recall the definition of $h(t)$ from (3.99) [p. 30]:

$$h(t) = f(t)e^{2a_{\min}t}. \tag{3.105}$$

By (3.105), we write (3.104) as

$$f(t)e^{2a_{\min}t} \leq h(0). \tag{3.106}$$

By (3.105), we also obtain $h(0)$:

$$h(0) = f(0)e^{2a_{\min}\cdot 0} = f(0). \tag{3.107}$$

Substituting (3.107) into (3.106),

$$f(t)e^{2a_{\min}t} \leq f(0). \tag{3.108}$$

Because $e^{2a_{\min}t} > 0$, dividing (3.108) by $e^{2a_{\min}t}$ gives

$$f(t) \leq f(0)e^{-2a_{\min}t}. \tag{3.109}$$

This reveals another reason we defined $h(t)$ as in (3.99) [p. 30]; $e^{2a_{\min}t}$ is positive and grows exponentially, so dividing by it gives an exponentially decaying function without changing the inequality's sign. Hence, $f(t)$ is bounded above by a function that tends to zero as $t \to \infty$.

Recalling the definition of $f(t)$ from (3.55) [p. 22],

$$f(t) = \mu_1 s_1^2 + \mu_2 s_2^2 + \cdots + \mu_k s_k^2, \tag{3.110}$$

where $\mu_1 = 1$ and $\mu_{n+1} = \mu_n c_n / b_{n+1}$ for $1 \leq n \leq k-1$, each term is nonnegative (justified in Remark 3.9 [p. 23]). Therefore,

$$0 \leq f(t) \leq f(0)\mathrm{e}^{-2a_{\min}t}.$$

Because $a_{\min} > 0$ from (3.100) [p. 30], we have

$$\lim_{t\to\infty} \mathrm{e}^{-2a_{\min}t} = 0,$$

it follows that

$$\lim_{t\to\infty} f(t) = 0.$$

Using the definition:

$$f(t) = \sum_{n=1}^{k} \mu_n s_n(t)^2, \tag{3.111}$$

where $\mu_1 = 1$ and $\mu_{n+1} = \mu_n c_n / b_{n+1}$ for $1 \leq n \leq k-1$, each term $\mu_n s_n(t)^2$ is nonnegative, so

$$\lim_{t\to\infty} \mu_n s_n(t)^2 = 0.$$

Since $\mu_n > 0$,

$$\lim_{t\to\infty} s_n(t)^2 = 0,$$

therefore

$$\lim_{t\to\infty} s_n(t) = 0.$$

Finally, from Definition 3.7 [p. 19],

$$P_n(t) = s_n(t) + P_n^*,$$

it follows that

$$\lim_{t\to\infty} P_n(t) = P_n^*.$$

□

We have proved that all populations tend to their equilibria as $t \to \infty$.

3.5 Summary

In this chapter,

1. We proved that the system has a unique equilibrium (Section 3.1 [p. 7]).
2. We used numerical simulations to motivate a conjecture that the system converges to equilibrium as $t \to \infty$ (Section 3.2 [p. 18]).
3. We performed a change of variables to define a system of displacements from equilibrium to simplify the following proof (Section 3.3 [p. 19]).
4. We proved that the system converges to equilibrium as $t \to \infty$ (Section 3.4 [p. 22]).

Chapter 4

Discussion and Evaluation

In Chapter 3 [p. 7], we showed that the system

- has a unique equilibrium $(P_1^*, \ldots, P_k^*)$;
- converges to equilibrium as $t \to \infty$.

Thus, we now know the system's long-term behaviour, helping answer our research question:

How effectively does a linear system of first-order ordinary differential equations model the long-term behaviour of a multi-trophic-level ecosystem subject to a pulse disturbance?

However, to answer our research question, we must evaluate our results' effectiveness. We therefore compare them with real-world observations and evaluate the model.

4.1 Discussion

The model's long-term predictions are supported by the literature, although they remain contested. The main alternatives are:

- **Return to the pre-disturbance equilibrium:** The system returns to its original stable equilibrium.
- **Persistent oscillations:** The system exhibits sustained oscillations rather than converging to equilibrium.
- **Regime shifts:** The system transitions to a different stable equilibrium.

A 2017 study by Karakoç et al. [2] found that, following a pulse disturbance, the predator recovered to its pre-disturbance status, while the prey remained slightly elevated.

These findings are broadly consistent with our model, which predicts convergence to the pre-disturbance equilibrium. The authors attributed the prey's incomplete recovery to adaptation, highlighting a limitation of our model's constant parameters; allowing coefficients to vary over time could capture such adaptation.

However, a 2020 study by Blasius et al. [3] observed persistent *predator–prey oscillations* over approximately 50 cycles. Predator–prey oscillations are cyclical population changes in predator and prey species. This behaviour is also visible in our numerical simulation (Figure 3.1 [p. 18]). Using plankton with short reproductive cycles, the authors obtained much longer datasets than typically available, as field observations usually span only a few cycles and may require decades to collect. They argued that these observations provide empirical evidence for potentially indefinite cyclic persistence, contrasting with the equilibrium behaviour predicted by our model. The study reported occasional transitions from regular, coherent oscillations to irregular, non-coherent oscillations, attributed to random fluctuations. Since our model is deterministic, it cannot reproduce such fluctuations, highlighting an important limitation. A further limitation is not proving the existence of oscillations under certain conditions, although our numerical simulations strongly suggest such behaviour. Predator–prey cycles are a well-known ecological phenomenon, and proving their existence would strengthen the model.

Regime shifts occur when a system has multiple stable equilibria, allowing a disturbance to move it from one equilibrium to another [4]. We have shown that our system has a unique stable equilibrium, so regime shifts are impossible. However, a critical review by Capon et al. [5] found that empirical evidence for regime shifts in freshwater ecosystems is generally weak. Thus, our model's unique stable equilibrium may be reasonable for some ecosystems. However, populations need not converge to an equilibrium, as illustrated by the persistent oscillations observed by Blasius et al. [3].

Overall, the literature suggests that our model's long-term predictions may be reasonable for some ecosystems, but not others.

4.2 Strengths

The model applies generally to food chains with $k \geq 2$ trophic levels. This is a strength because real-world ecosystems often have multiple trophic levels, unlike traditional models focusing on two. Furthermore, its qualitative behaviour is proved analytically without additional assumptions or parameter restrictions. This is advantageous over relying on numerical simulations, whose results depend on specific parameter values and initial conditions. Since model parameters are abstractions of real-world quantities, often difficult

to determine precisely, numerical predictions may be incorrect due to measurement errors. Our long-term predictions are a strength, as they hold independently of specific parameter values (beyond the basic restrictions of the model) and initial conditions.

4.3 Limitations

4.3.1 Linear Interactions

Our model uses $-b_n P_{n-1}$ to represent predation by the higher trophic level and $c_n P_{n+1}$ to represent prey's contribution to population growth. A more realistic model would use nonlinear interaction terms (e.g., $-b_n P_{n-1} P_n$), since predation depends on both predator and prey populations. For example, if $P_2 = 0$, the term $c_2 P_3$ can increase P_2, implying that an extinct population can recover solely because prey are present. This is biologically unrealistic. A nonlinear term such as $c_2 P_2 P_3$ instead vanishes when $P_2 = 0$, preventing this. This limitation suggests the model is most appropriate when populations remain nonnegative, or when disturbances Δ_n, $n \in \{1, \ldots, k\}$, are small enough to avoid extinction or negative populations.

4.3.2 Lack of Stochasticity

Stochasticity means incorporating randomness into a model [6]. Our model is deterministic, so its behaviour is determined by its initial conditions and parameters. Ecological systems experience random environmental fluctuations that can accumulate and significantly affect their behaviour. However, stochasticity would substantially increase model complexity. Nevertheless, Blasius et al. [3] found that, despite stochasticity, the system still tended towards stable oscillations, suggesting that a deterministic model (not necessarily ours) can be a reasonable approximation.

4.3.3 Constant Coefficients

Our model cannot account for adaptation or other time-dependent changes because its parameters are constant. This is a limitation, as adaptation to disturbances may affect long-term behaviour. However, such adaptation is likely species- and ecosystem-specific, so time-dependent coefficients would reduce the model's generality. Furthermore, Karakoç et al. [2] found that the predator returned to its pre-disturbance level, while the prey

remained slightly elevated. This suggests that adaptation may not be significant in some ecosystems, so a model with constant coefficients may still be reasonable.

4.3.4 Nonnegativity

We did not derive the exact parameter conditions ensuring nonnegative populations. Instead, numerical simulations only show that nonnegative populations appear possible over part of the domain (Figure 3.1 [p. 18]); this is not a proof. We omitted this derivation to avoid substantial algebraic manipulation and keep the essay's length manageable.

4.3.5 Food Webs

Recall our model from (3.4) [p. 8]:

$$\frac{dP_n}{dt} = \begin{cases} \lambda_1 - a_1P_1 + c_1P_2, & n = 1 \\ \lambda_n - b_nP_{n-1} - a_nP_n + c_nP_{n+1}, & 2 \leq n \leq k-1, \\ \lambda_k - b_kP_{k-1} - a_kP_k, & n = k. \end{cases} \tag{4.1}$$

In reality, ecosystems form *food webs* rather than food chains [7]. For example, P_3 may be consumed by both P_1 and P_2, while consuming both P_4 and P_5.

Our model, adapted as a food web with $k = 4$ trophic levels, is

$$\frac{dP_n}{dt} = \begin{cases} \lambda_1 - a_1P_1 + c_{12}P_2 + c_{13}P_3 + c_{14}P_4, & n = 1, \\ \lambda_2 - b_{21}P_1 - a_2P_2 + c_{23}P_3 + c_{24}P_4, & n = 2, \\ \lambda_3 - b_{31}P_1 - b_{32}P_2 - a_3P_3 + c_{34}P_4, & n = 3, \\ \lambda_4 - b_{41}P_1 - b_{42}P_2 - b_{43}P_3 - a_4P_4, & n = 4. \end{cases} \tag{4.2}$$

The parameters have the same interpretation as in the food chain model, except that c_{ij} denotes the growth of P_i from consuming P_j, while b_{ij} denotes the death of P_i due to predation by P_j.

An equilibrium can be found by Gaussian elimination under suitable parameter conditions. We may also use displacements to analyse the system as for the food chain model, but this would require substantial algebraic manipulation. Doing so would likely reduce clarity.

Nevertheless, this extension could increase the model's realism without greatly increasing complexity, as it remains linear, making it a possible research direction.

4.4 Summary

We evaluated our results against real-world observations. We conclude that our model may capture the long-term behaviour of some ecosystems, albeit not all. We also discussed the model's strengths. Additionally, we discussed the model and essay's limitations. Many model limitations stem from its linearity. However, other limitations arise from mathematical choices throughout the essay. Most aimed to keep the essay manageable and minimise lengthy algebraic manipulation, which could have reduced clarity.

Chapter 5

Conclusion

This essay set out to answer: **How effectively does a linear system of first-order ordinary differential equations model the long-term behaviour of a multi-trophic-level ecosystem subject to a pulse disturbance?**

Based on our long-term result in Chapter 3 [p. 7] and its discussion, we conclude that it is applicable to some ecosystems. However, contradictory empirical evidence from Blasius et al. [3] suggests that it may not apply to all ecosystems. Thus, we conclude that the model is **partially effective** in modelling the long-term behaviour of multi-trophic-level ecosystems.

We first introduced a linear model of a food chain with $k \geq 2$ trophic levels in Chapter 2 [p. 2]. We then proved in Chapter 3 [p. 7] that the system has a unique equilibrium and converges to this equilibrium as $t \to \infty$. In Chapter 4 [p. 34], we compared this long-term result with empirical findings and evaluated the model. We concluded that its lack of stochasticity and use of constant coefficients may be less problematic than initially thought, while its linearity and omission of food webs limit its realism. Finally, we discussed limitations arising from our mathematical choices, such as not deriving nonnegativity conditions and not investigating a food-web model.

Ultimately, while the model is only partially effective, it demonstrates how a system can be generalised to k trophic levels by simplifying it through linear modelling. Some realism is inevitably lost; however, our linear model may still provide a reasonable approximation of some ecosystems and serves as a useful starting point for further research. Future work could investigate the long-term behaviour of a food-web model, increasing realism without greatly increasing complexity, since it remains linear.

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